PRACTICE SHEET

1. If $\int \sec x \operatorname{cosec} x\, dx = \log|g(x)| + c$, then what is g (x) equal to?
(a) sin x cos x (b) $\sec^2 x$
(c) tan x (d) log|tan|x

2. What is the value of $\int (\sqrt{x}+x)^{-1}$?
(a) $\ln(x+\sqrt{x})+c$ (b) $2\ln(1+\sqrt{x})+c$
(c) $2\ln(x+\sqrt{x})+c$ (d) $2\ln(x-\sqrt{x})+c$

3. What is the value of $\int \frac{e^x(1+x)}{\sin^2(xe^x)}dx$?
(a) $-e^x \cot x + c$ (b) cos2 (xe^x) + c
(c) log sin $(x\, e^x)$+c (d) $-\cot(x\, e^x) + c$

4. What is $\int \log(x+1)\, dx$ is equal to?
(a) x log (x + 1) − x + c
(b) (x + 1) log (x + 1) − x + c
(c) $\frac{1}{x+1}+c$
(d) $\frac{\log(x+1)}{x+1}+c$

5. If $\int \frac{dx}{f(x)} = \log\{f(x)\}^2 + c$, then what is (x) equal to?
(c) $\frac{\tan^5 x}{5}+\frac{\sec^3 x}{3}+c$ (d) $\frac{\sec^5 x}{5}+\frac{\tan^3 x}{3}+c$

10. What $\int \sec x^\circ\, dx$ is equal to?
(a) log $(\sec x^\circ + \tan x^\circ) + c$
(b) $\frac{\pi \log\tan\left(\frac{\pi}{4}+\frac{\pi}{2}\right)}{180^\circ}+c$
(c) $\frac{180^\circ \log\tan\left(\frac{\pi}{4}+\frac{\pi}{2}\right)}{\pi}+c$
(d) $\frac{180^\circ \log\tan\left(\frac{\pi}{4}+\frac{x}{360^\circ}\right)}{\pi}+c$

11. What is $\int \frac{a+b\sin x}{\cos^2 x}$ equal to?
(a) $a\sec x + b\tan x + c$
(b) $a\tan x + b\sec x + c$
(c) $a\cot x + b\operatorname{cosec} x + c$
(d) $a\operatorname{cosec} x + b\cot x + c$

12. What is $\int \frac{\log x}{(1+\log x)^2}dx$ equal to?
(a) $\frac{1}{(1+\log x)^3}+c$ (b) $\frac{1}{(1+\log x)^2}+c$